ANALOG & MIXED-SIGNAL VLSI · PROJECT DOCUMENTATION

10-bit SAR ADC in 90 nm CMOS

A successive-approximation analog-to-digital converter designed transistor by transistor in Cadence Virtuoso and verified in Spectre: sample-and-hold, dynamic comparator, binary-weighted capacitor DAC and a flip-flop-based SAR controller, integrated into one working converter.

10 bit resolution · 1 LSB ≈ 1.46 mV	**900 kS/s** sampling rate (1.11 µs per conversion)	**31.18 µW** total power in simulation
1.5 V single supply	**90 nm** CMOS · built from individual transistors	**1.053 µs** delay from sampling the input to a valid output word

My role	**SAR control logic and top-level integration:** the flip-flop-based SAR register and End-of-Conversion logic, connecting all four blocks into the full ADC, and the full-ADC transient and power simulations. Also the independent design review and roadmap in Sections 8–9
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Context	Academic project, B.E. Electronics & Communication, Dayananda Sagar College of Engineering, Bengaluru (course 22EC66)
Timeline	Academic year 2024–25 · report submitted May 2025
Tools	Cadence Virtuoso (schematic entry, Analog Design Environment) · Cadence Spectre (transient simulation and power measurement) · 90 nm CMOS device models
Scope	Schematic-level design and simulation. Layout and fabrication were out of scope (see Section 8)

1 Summary

An ADC converts a real-world signal, such as a heartbeat, a temperature or a vibration, into numbers a processor can work with. SAR ADCs are the standard choice when moderate resolution (8–12 bits) at a moderate sample rate is needed on a tight power budget: they need almost no amplification circuitry, and most of the work is done by a single comparator and a bank of switched capacitors.

I looked at recent published 10-bit SAR ADCs. Each was very good on one measure but weak on another: very low power but slow (500 kS/s, or even 1 kS/s), or faster but power-hungry (90 µW). This project set three targets together:

SPEED

At least **900 kS/s**, which means a full 10-bit conversion every 1.11 µs.

POWER

Less than **90.11 µW**, the power of the fastest comparable design.

SUPPLY

Stable operation from a single **1.5 V** supply in 90 nm CMOS.

> **Result:** the integrated ADC produced correct 10-bit output words for a sine-wave input in transient simulation, at **900 kS/s** and **31.18 µW** from 1.5 V. That is about one third of the power budget, at a faster sample rate than the low-power designs it was compared against.

Where the design sits among published designs

Metric	Huang & Wu, 2025	Kumaradasan et al., 2024	Moghadam et al., 2024	This design
Technology	40 nm	180 nm	65 nm	**90 nm**
Resolution	10 bit	10 bit	10 bit	**10 bit**
Sampling rate	500 kS/s	833.33 kS/s	1 kS/s	**900 kS/s**
Power	0.9047 µW	90.11 µW	2.08 nW	**31.18 µW**
Supply	0.7 V	1.6 V	0.2 V	**1.5 V**
Best suited to	Low speed, low power	High speed, higher power	Ultra-low power, very slow	**Balanced: real-time capable**

The published designs report measured or post-layout figures; this design's figures come from schematic-level simulation. Section 8 explains what that means for a fair comparison.

2 Architecture

A SAR ADC finds the digital code for the input with a binary search: it first decides the most significant bit (MSB), then each lower bit in turn. Each bit takes one comparator decision, so a 10-bit conversion takes 10 clock cycles after the input is sampled.

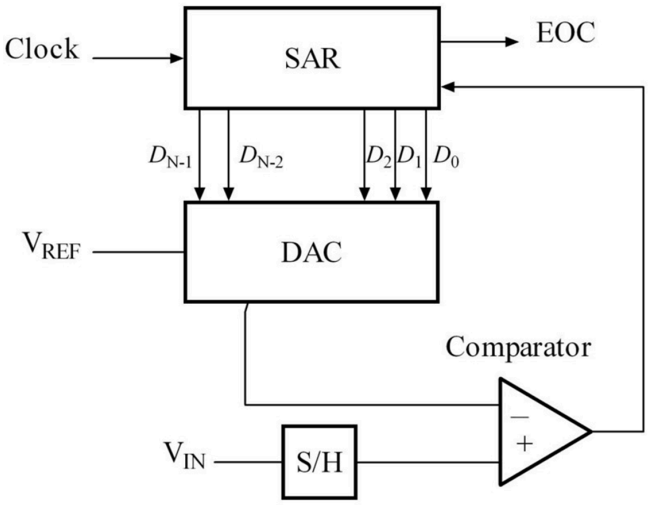


Figure 1. SAR ADC block diagram.

Block	What it does
S/H	Captures the input voltage and holds it steady while all 10 bits are decided
Comparator	Decides whether the held input is above or below the DAC voltage, producing 1 bit per clock
CDAC	Converts the current trial code back into a voltage, using binary-weighted capacitors
SAR logic mine	Sets each trial bit, keeps or clears it based on the comparator result, stores the result, and signals End-of-Conversion (EOC)

The successive-approximation algorithm

```
sample V_IN on the hold capacitor;  code = 0
for k = 9 down to 0:                 // MSB first, one clock cycle per bit
    code[k] = 1                      // trial: set this bit
    V_DAC   = code · V_REF / 2^10    // CDAC produces the trial voltage
    if V_IN < V_DAC:  code[k] = 0    // comparator says too high, so clear the bit
assert EOC;  output code[9:0]
```

Each step halves the range the answer can be in, so after 10 decisions the error is under 1 LSB ($V_{REF}/1024 \approx 1.46$ mV at 1.5 V). Because the logic is a fixed, clocked sequence, power stays low: the circuit only draws current when the comparator fires and when capacitors switch.

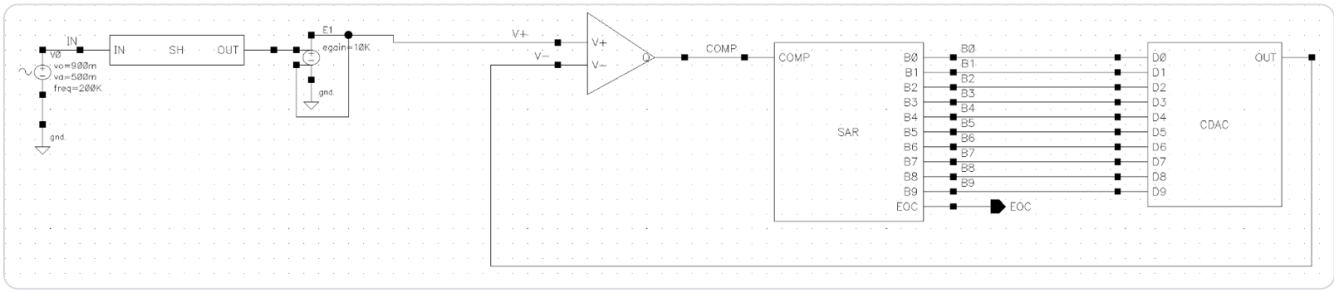


Figure 2. Top-level schematic of the integrated ADC in Cadence Virtuoso: S/H, comparator, SAR logic and CDAC connected in the feedback loop. Building and verifying this integration was my part of the work.

3 The analog blocks

Sample-and-hold

- **Transmission-gate switch** (an NMOS and PMOS in parallel, W/L = 500 nm / 100 nm) so the switch conducts well across the whole input range.
- **Switch resistance:** $R_{on,n} \approx 1.40$ kΩ and $R_{on,p} \approx 3.74$ kΩ, about 1.0 kΩ in parallel.
- **Sampling capacitor:** $C_S \approx 178$ fF, giving a sub-nanosecond RC time constant.
- **Sampling pulse:** 1.11 µs period, 100 ns sampling window, 1 ps edges.
- **Test input:** 0.9 V + 0.5 V·sin(2π·100 kHz·t).

The 100 ns sampling window is many times longer than the RC time constant, so the capacitor charges fully to the input before each conversion (checked in Section 8).

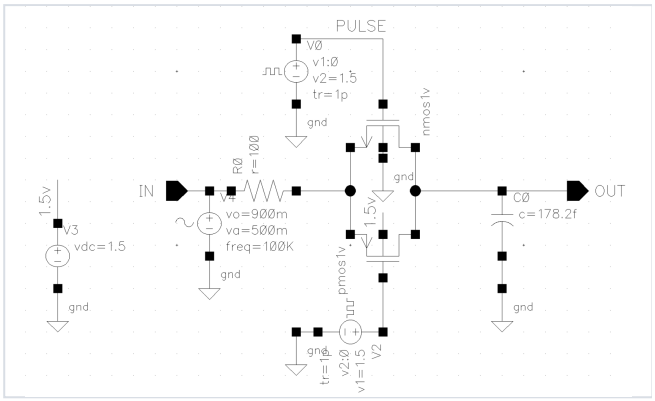


Figure 4. S/H schematic: transmission gate and 178 fF hold capacitor.

Dynamic latched comparator

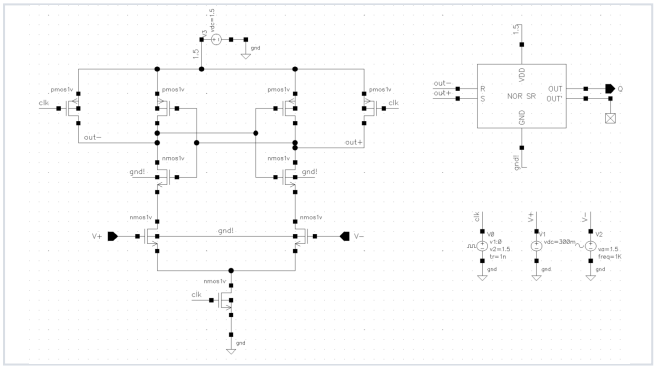


Figure 5. Clocked comparator with a cross-coupled latch.

- **Clocked, cross-coupled inverter latch** (the StrongARM style of comparator): it amplifies a small input difference until the output snaps to a full logic 1 or 0.
- **No static current:** it only draws current when it makes a decision, the main reason the total power is so low.
- **Output COMP** goes straight to the SAR logic as that cycle's bit decision.
- **Checked on its own** with a 300 mV DC reference against a time-varying input, clocked at 10 µs: COMP is high exactly when V+ > V−.

Capacitive DAC (CDAC)

- **Binary-weighted array:** 512C, 256C, ..., 2C, C plus one dummy C, with a unit capacitor C = 2.5 fF, so the total array is 1024C ≈ 2.56 pF.
- **Switching:** each bit B[k] drives a transmission gate that connects its capacitor to the reference or to ground. Charge sharing across the array produces $V_{OUT} = V_{REF} \cdot (\text{code} / 1024)$.
- **Dummy capacitor:** makes the total exactly 2^{10}·C, so each bit's weight is an exact power of two.

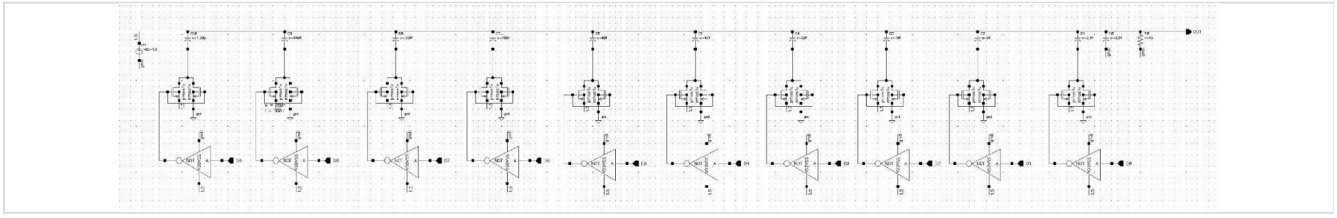


Figure 6. The 10-bit binary-weighted CDAC with transmission-gate switches.

4 SAR logic my block

I designed the SAR logic, which controls the whole conversion. On each clock it does three things: sets the next bit to try, keeps or clears the previous bit depending on the comparator, and stores the result. After the last bit it signals End-of-Conversion.

STRUCTURE

- **Sequence register:** a chain of D flip-flops that passes a single 1 along, from B9 down to B0, choosing which bit is being tried in each cycle.
- **Code register:** ten flip-flops that store B[9:0]. When it is a bit's turn, its flip-flop is set to 1; one clock later it loads COMP, which keeps the 1 if the input was higher and clears it if lower.
- **EOC:** generated when the last decision (the LSB) is made, telling the next stage the data is ready.
- **Reset:** a 20 ns active-low pulse at the start of each 1.11 µs conversion puts the register back to `1000000000`.
- **My own D flip-flop:** built from transistors so that I controlled its timing and power. It is used in every register position.

INTERFACE AND TEST STIMULUS

- **Inputs:** CLK (110 ns period, 55 ns high), RESET (20 ns pulse every 1.11 µs), COMP, SET, VDD = 1.5 V.
- **Outputs:** B[9:0], EOC.
- **Tested alone first:** I drove COMP with a known pattern (1.11 µs period, 600 ns high), so the expected output word was known in advance and the logic could be checked on its own before integration.

What happens in each cycle

Cycle	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0	COMP	EOC
Initial	**1**	0	0	0	0	0	0	0	0	0	A9	0
1	A9	**1**	0	0	0	0	0	0	0	0	A8	0
2	A9	A8	**1**	0	0	0	0	0	0	0	A7	0
3	A9	A8	A7	**1**	0	0	0	0	0	0	A6	0
…	a decided bit stays fixed, and the trial 1 moves one place towards the LSB										…	0
9	A9	A8	A7	A6	A5	A4	A3	A2	A1	**1**	A0	0
10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0	–	**1**

A_k is the comparator's decision for bit k. The bold 1 is the bit being tried in that cycle.

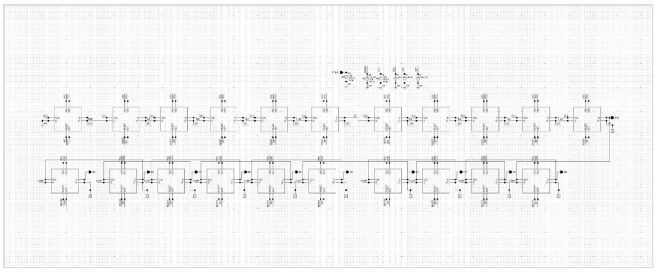


Figure 7. SAR logic schematic: sequence register and code register.

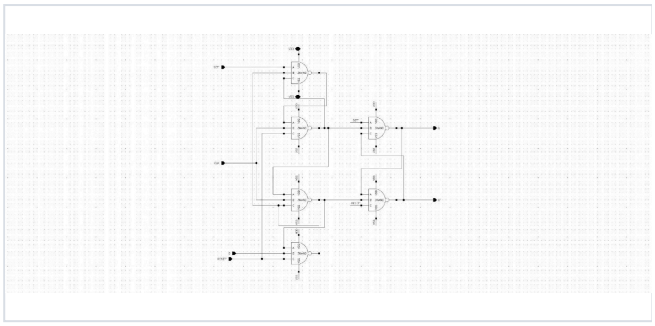


Figure 8. The transistor-level D flip-flop used throughout the SAR logic.

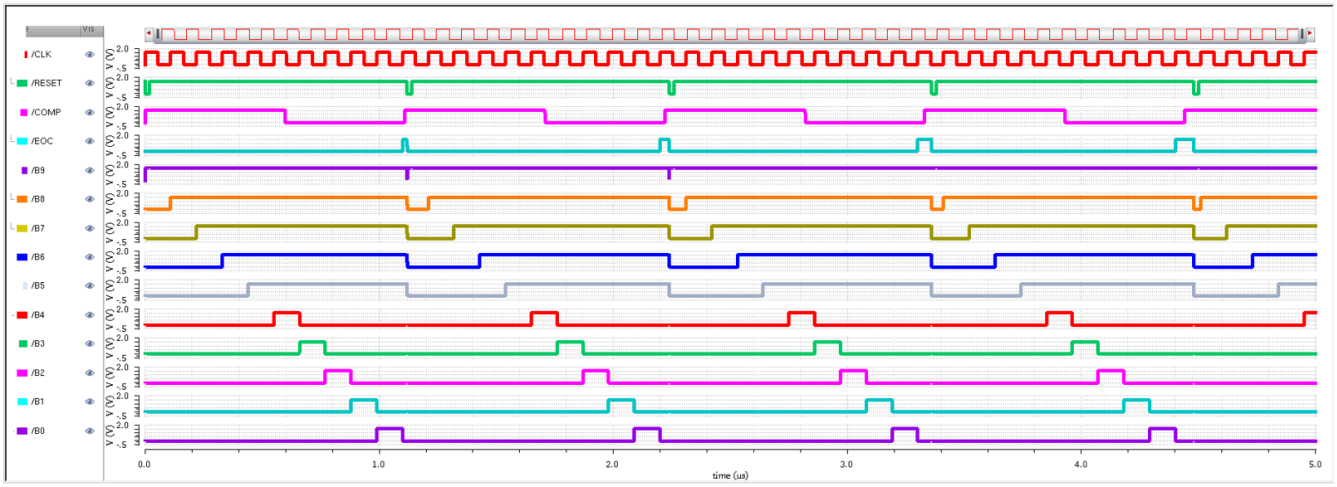
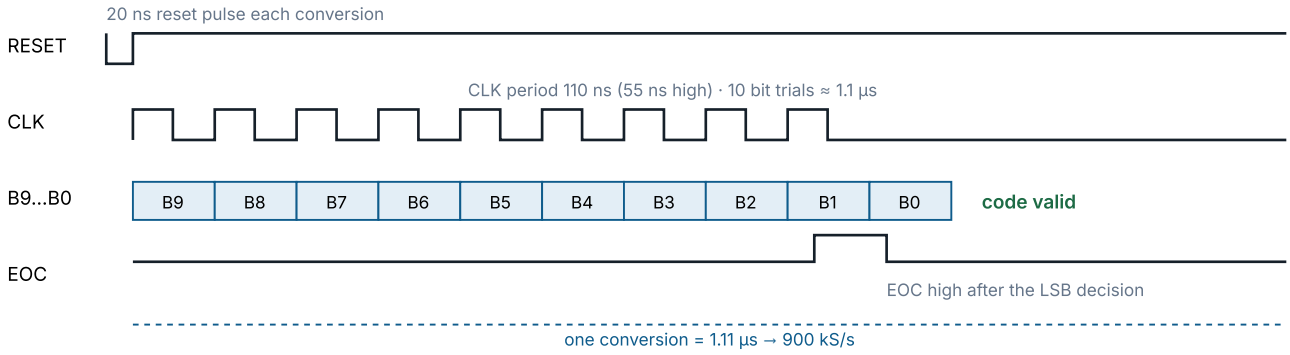


Figure 9. SAR logic simulation: CLK, RESET and COMP stimulus with B9–B0 resolving one at a time and EOC going high at the end of each conversion.

Timing for one conversion



Conversion timing, from the simulation stimulus: a reset pulse starts it, there is one bit trial per clock from MSB to LSB, and EOC goes high when the 10-bit word is ready.

5 Integration and full-ADC verification my work

The four blocks were each verified first, then I connected them into the complete loop (S/H → comparator → SAR logic → CDAC → back to the comparator) and simulated the whole ADC with a sine-wave input.

HOW I VERIFIED IT

1. **Each block on its own**, with a transient simulation and stimulus that gave a known expected output (Section 6).
2. **Interfaces:** logic levels between the blocks, the comparator output reaching the SAR logic within the 55 ns high phase of the clock, and the CDAC voltage settling before each comparator decision.
3. **Full ADC:** a sine input across the 0–1.5 V range, sampled at 900 kS/s, with the output word checked against the input level at several points in time.
4. **Power and delay** measured in the ADE (Analog Design Environment) outputs: **31.1782 µW** total power, **1.05314 µs** from sampling to output.

INTEGRATION CONSIDERATIONS

- **Timing between blocks:** the 110 ns clock (10 bit trials = 1.10 µs) had to fit inside the 1.11 µs sampling period. That leaves just one 10 ns slot for reset and sampling, so the RESET pulse and the sampling window had to line up exactly.
- **Waiting for the CDAC:** each comparator decision has to happen after the CDAC voltage has settled, so the comparator fires on the clock edge after the bit changes.
- **Simulation hierarchy:** each block's symbol and test bench was built so the same cell could be simulated alone or as part of the full ADC without changes.

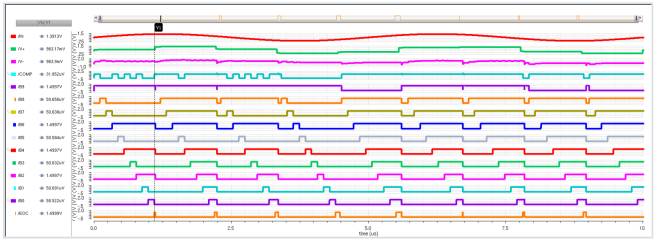


Figure 10. Full-ADC transient simulation: analog input with the B9–B0 output bits and EOC. The output code follows the input level.

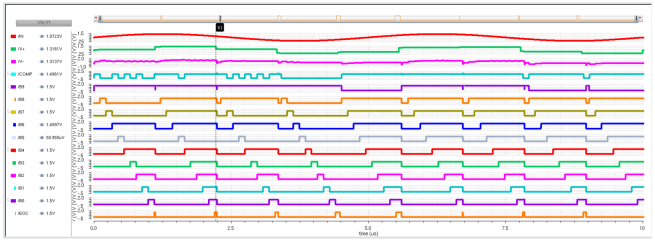


Figure 11. The same simulation at a later time: the output code changes as the sine input sweeps through its range.

6 Block-level simulation results

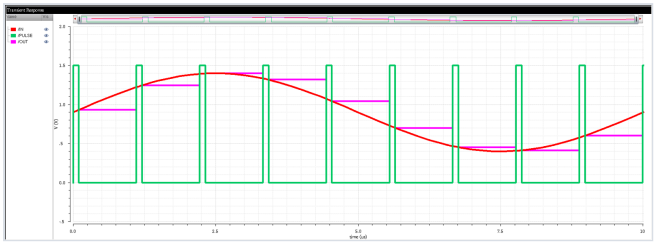


Figure 12. S/H: the output steps follow the 100 kHz sine and stay flat between samples.

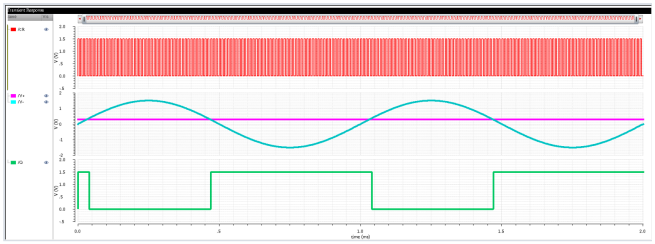


Figure 13. Comparator: the output is high while V+ > V− and changes cleanly on each clock.

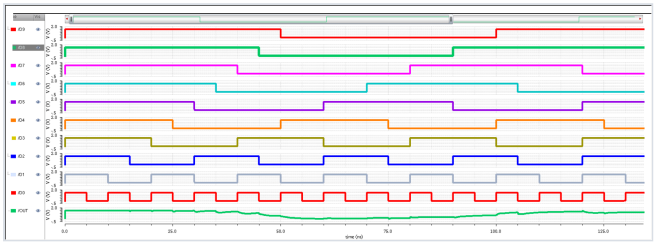


Figure 14. CDAC: digital inputs D9–D0 and the matching analog output steps.

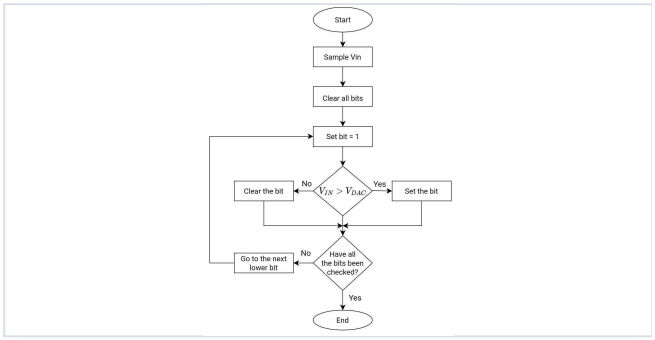


Figure 15. Conversion control flow: sample, clear, then set and test each bit from MSB to LSB, then finish.

Performance summary (schematic-level simulation)

Parameter	Target	Result	Status
Resolution	10 bit	10-bit output word, B[9:0]	Met
Sampling rate	≥ 900 kS/s	900 kS/s (1.11 µs per conversion)	Met
Power	< 90.11 µW	31.18 µW	Met (65% below the target)
Supply	1.5 V	1.5 V	Met
Delay from sampling to output	< 1.11 µs	1.053 µs	Met

7 What I learned

DESIGN

- Designing digital control logic out of transistors, where every gate's timing affects the analog loop around it.
- Fitting several blocks into a fixed timing budget: 10 bit trials, a reset and sampling, all within 1.11 µs.
- The SAR trade-off: resolution versus conversion time versus switching energy.

TOOLS AND METHOD

- Cadence Virtuoso hierarchical schematics, symbols and test benches.
- Spectre transient simulation, and power and delay measurements in ADE.
- Verifying each block on its own before integrating, with a known stimulus and a known expected output.

8 Design review: the numbers checked independent

I went back over the design and checked its key numbers by hand. This section records what holds up, what needs correcting, and what a fair comparison with published designs has to take into account.

1 · Does the S/H settle in time? Yes, with a large margin

Check $R_{on} \parallel = (1402.9 \cdot 3738.3)/(1402.9 + 3738.3) \approx$ **1.02 kΩ** (the report's 1.12 kΩ is a small arithmetic slip). $\tau = 1.02\ k\Omega \times 178\ fF \approx$ **0.18 ns**. Settling to within ½ LSB at 10 bits takes $\ln(2^{11}) \approx 7.6\ \tau \approx$ **1.4 ns**.

Conclusion The 100 ns sampling window is about 70 times longer than needed. Even ADE's measured time constant of 1.58 ns (which includes source resistance) settles in about 12 ns. The capacitor could be made larger, which lowers noise, without hurting speed.

2 · Is sampling noise acceptable? Yes

Check Sampling noise $\sqrt{(kT/C)} = \sqrt{(1.38 \cdot 10^{-23} \cdot 300 / 178 \cdot 10^{-15})} \approx$ **152 µV rms**, against 1 LSB = 1.5 V/1024 ≈ **1.46 mV**.

Conclusion About 0.1 LSB rms, small enough for 10 bits. A 2 pF capacitor would bring it down to about 45 µV at almost no speed cost (see item 1).

3 · Energy efficiency: a fairer comparison

Check The Walden figure of merit, $FoM = P / (2^{ENOB} \cdot f_s)$, measures energy per conversion step. Assuming an ideal 10-bit ENOB (effective number of bits): 31.18 µW / (1024 × 900 kS/s) ≈ **33.8 fJ/step**. This is a best case: a real ENOB below 10 makes the figure higher.

Conclusion For comparison, Kumaradasan et al. are about 137 fJ/step (ENOB ≈ 9.6 from their 59.7 dB SNDR) and Huang & Wu about 2.8 fJ/step. So this design is clearly more efficient than the 180 nm design, but not yet at the level of the best 40 nm designs. Measuring the real ENOB (Section 9) is what would make this comparison final.

4 · CDAC capacitor mismatch: the main risk in silicon

Check A 2.5 fF unit capacitor is about the smallest that can realistically be built. In a binary array, the MSB capacitor (512C) is matched against the sum of all the others, and the error at the MSB transition, the middle of the range, sets the worst-case DNL (differential nonlinearity). The simulation used ideal capacitors.

Conclusion Linearity has to be checked with Monte Carlo simulation using real mismatch models, before trusting the 10-bit claim on silicon.

5 · What has not been characterised yet

Gap The design was checked by looking at transient output codes only. There are no DNL/INL (linearity), SNDR/ENOB (dynamic accuracy) or PVT corner results yet. Its "10-bit" status is therefore functional, not yet a measured accuracy. The published designs it is compared against report post-layout or silicon results.

9 Roadmap: from schematic to silicon Proposed

1 · CHARACTERISATION

- **Static linearity:** feed in a slow ramp (or a sine) and count how often each output code appears. From that histogram, compute DNL and INL across all 1024 codes.
- **Dynamic accuracy:** use coherent sampling ($f_{in} = (M/N) \cdot f_s$, with M odd and prime to N), take an FFT of the output, and calculate SNDR, SFDR and ENOB, which give the real FoM.
- **Robustness:** repeat across process, voltage and temperature corners (SS/FF/SF/FS, ±10% supply, −40 to 125 °C) and run Monte Carlo on the CDAC and comparator offset.

2 · CIRCUIT IMPROVEMENTS

- **Monotonic or split-capacitor CDAC switching:** large reduction in switching energy and total capacitance compared with conventional switching.
- **Bootstrapped sampling switch:** keeps the switch resistance constant across the input range, improving linearity at high input frequencies.
- **Comparator offset calibration** at power-up.
- **Asynchronous SAR timing:** each bit trial starts as soon as the comparator has decided, instead of waiting for a fixed clock. This removes the fast external clock and increases speed.

3 · PHYSICAL DESIGN

- **Common-centroid CDAC layout** with dummy capacitors around the array, so manufacturing variation affects all capacitors equally.
- **Shielding** of the comparator inputs from the digital switching nodes.
- **DRC/LVS-clean layout**, then post-layout parasitic extraction and simulation, comparing against the schematic results.

4 · PRODUCTISATION

- **Selectable resolution** (8/10-bit) for different applications.
- **Digital calibration** of capacitor weights to correct mismatch.
- **A simple serial or parallel output** with EOC handshake, so the ADC can be connected directly to a microcontroller or DSP block.

Where this ADC fits

Wearable and implantable medical sensing (ECG, SpO_2), industrial sensor nodes (pressure, temperature, vibration), battery-powered IoT front ends, and any system that needs about 10 bits at a few hundred kS/s within a microwatt-level power budget.

Built in 2024–25 as part of a 4-member academic mini-project team under the guidance of Dr. Abhishek M B, Dept. of ECE, DSCE, Bengaluru. The author's work: the SAR logic, top-level integration and full-ADC verification, and the independent review and roadmap in Sections 8–9. Comparison figures are taken from the cited publications (Huang & Wu, IEICE Electronics Express 2025; Kumaradasan et al., IEEE Xplore 2024; Moghadam et al., Arabian Journal for Science and Engineering 2024). The figure of merit for the published designs was calculated from their reported numbers.